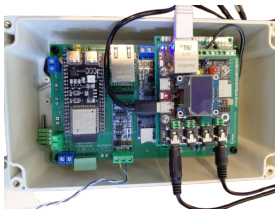
DTM-EMS

Transformer-edge intelligence for feeder-wide microgrid coordination, front and behind the meter





Outdoor IP66 enclosure

DTM-EMS is the intelligent controller at the heart of the secondary distribution transformer. It fuses transformer-grade polyphase precision metering (±0.1%) with neutral, phase and leakage insight, autonomous phase balancing and dynamic energy-policy enforcement, then **networks with multiple open-source Street-EMS nodes on a shared LV feeder** to coordinate the whole low-voltage network. Built on **open standards (IEEE, ISV, OpenAMI)**, it is the open, future-proof brain of the grid edge.

What Sets DTM-EMS Apart:

- **Precision metering + leakage insight** : ±0.1% polyphase, with neutral, phase and AC/DC leakage measured directly (plus NILM).
- **Autonomous phase balancing** : drives neutral current toward zero and cuts feeder losses, hands-off.
- **Feeder-wide coordination** : networks open-source Street-EMS nodes on one LV feeder for front-of-meter rural and urban village microgrids.
- **Front and behind the meter** : orchestrates behind-meter building EV charge / discharge and riser / balcony solar-and-battery.
- **Open-Source Hardware** : released under CERN-OHL-S v2 — schematics, board files and firmware published on GitHub (meshEMS / X-EMS).
- **Field-Serviceable** : built from open, off-the-shelf commodity modules, snap-in mounted for fast in-country assembly and repair.

One open, outdoor-rated box turns the distribution transformer into the coordinating brain of a feeder-wide microgrid, on standards you own.

Intelligent

- Neutral / phase / leakage + NILM insights
- Polyphase metering, ±0.1% active energy
- Autonomous phase balancing (neutral minimisation)

Interoperable

- Networks Street-EMS nodes on a shared LV feeder
- Open standards: IEEE / ISV / OpenAMI
- Ethernet / RS-485 / WiFi 6 / Thread / Zigbee / MQTT

Reliable

- Proven off-the-shelf metering hardware
- Onboard RTC, temperature & EEPROM | stackable ×64

Typical Applications

- Front-of-meter rural and urban village microgrid electrification
- Feeder-wide coordination of open-source Street-EMS nodes on a shared LV transformer
- Behind-the-meter building EV charge / discharge and riser / balcony solar-and-battery
- Neutral-current reduction, phase balancing and Type-B (AC+DC) leakage safety

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Product Specifications

PRODUCT	
Product name	DTM-EMS (10Power X-EMS, supplied by NESL)
Metering platform	Off-the-shelf polyphase precision-metering platform
Firmware	meshEMS-integrated open-source insights & policy layer
License & source	Hardware: CERN-OHL-S v2 | Repository: GitHub, meshEMS / X-EMS
Serviceability	Open, off-the-shelf commodity modules | snap-in mounted for fast in-country assembly and repair
METROLOGY	
Metering IC	Polyphase energy-metering IC
Channels	3 voltage inputs | up to 4 CT: 3 phase + 1 neutral (clamp or Rogowski)
Accuracy	±0.1% active energy | ±0.2% reactive energy
High-current input	Rogowski-coil support for high-amperage feeders
INSIGHTS	
Insights engine	Neutral | phase imbalance | leakage | NILM (non-intrusive load monitoring)
Measured	RMS V / I | P / Q / S | harmonics | PF | phase angle | frequency | bidirectional energy
POLICY ENFORCEMENT	
Edge enforcement	Dynamic energy-policy enforcement at the transformer edge
Phase balancing	Autonomous phase rebalancing to minimise neutral current and feeder losses
Feeder coordination	Networks with multiple open-source Street-EMS nodes on a shared LV feeder
INTEGRATION & NETWORKING	
Co-processor link	SPI + 40-pin expansion header (any MCU / Pi, stackable ×64)
Networking	Ethernet (RJ45) | RS-485 Modbus | WiFi 6 / Thread / Zigbee / Matter
Telemetry	MQTT | OpenAMI-compatible edge telemetry
Onboard	RTC | temperature sensor | EEPROM | RGB status
POWER & MECHANICAL	
Power supply	5 V USB-C or 4.5-20 V DC | or 8-12 V AC (SELV)
Dimensions	85 × 55 mm | or Raspberry Pi HAT form factor
Enclosure	Outdoor IP66 subpanel enclosure | converged EMS + DTM in one box
Mounting	Wall or DIN-rail | outdoor mains subpanel
STANDARDS & POSITIONING	
Grid position	Secondary distribution transformer (DTM), coordinating the LV feeder
Open standards	IEEE | ISV | OpenAMI compatible
Deployment	Front-of-meter rural / urban village microgrids | behind-meter EV charge / discharge | riser / balcony solar + battery
Capabilities	Polyphase precision metering | neutral / phase / leakage insight | autonomous phase balancing | feeder-wide coordination

Key Benefits

- Transformer-grade visibility: neutral, phase, AC/DC leakage and harmonics in one device
- Coordinates a whole LV feeder of open-source Street-EMS nodes, front and behind the meter
- Enables rural and urban village microgrid electrification and behind-meter EV, solar and battery
- Open standards (IEEE / ISV / OpenAMI): future-proof, with no vendor lock-in